HabbitQuest AI Deep Dive

Nov 3, 2026 · @Rahul

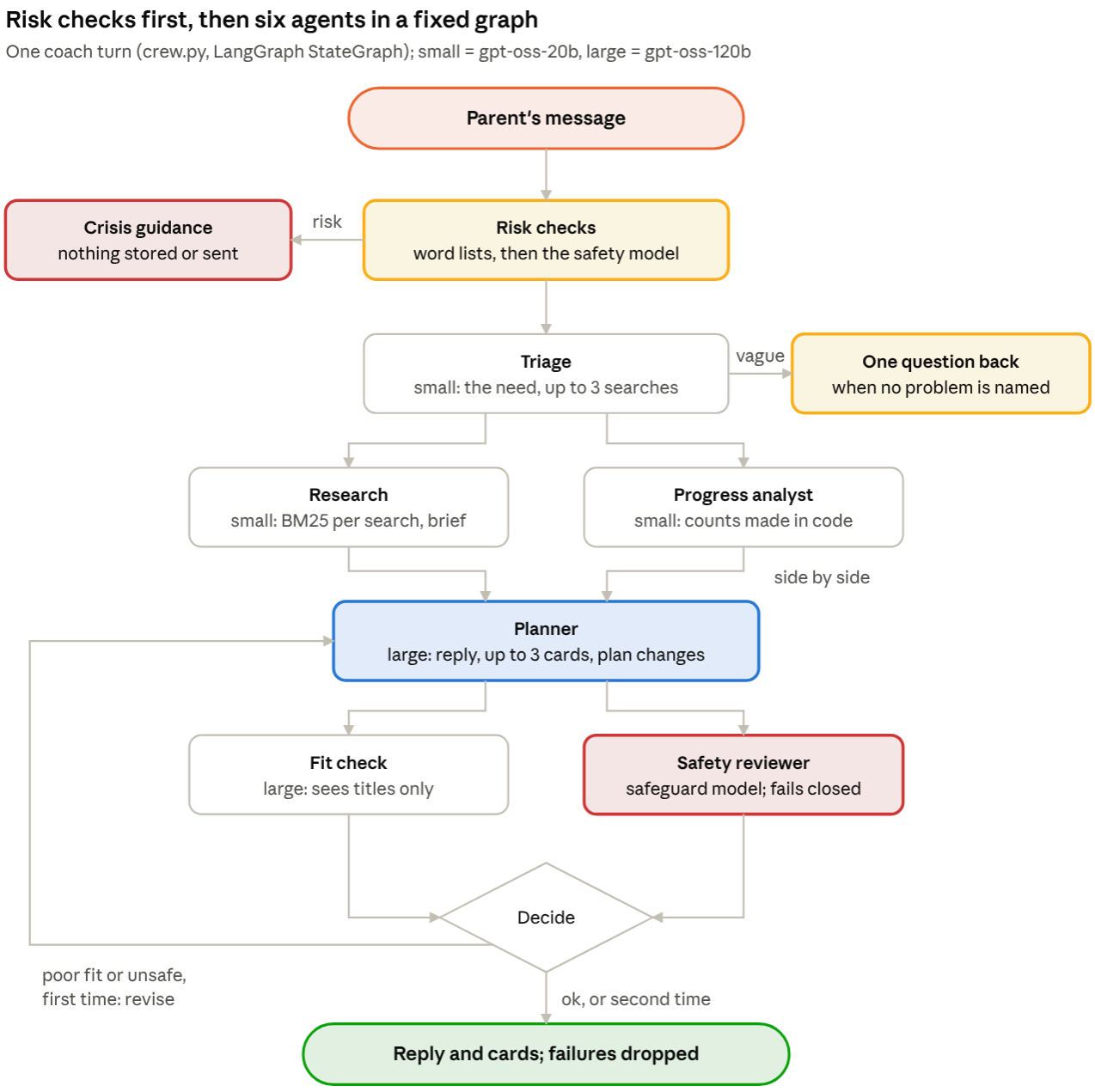
The habit coach is an agentic crew of six small agents on LangGraph, wrapped in safety checks before and after every model call. It passes 37 of 41 realistic test cases, and its runs can be traced in LangSmith with personal details removed.

1. The AI features

Every AI feature only suggests. A grown-up accepts each suggestion, and the server checks the plan rules again.

Feature	What it does	Models (Groq)	Pattern
Habit coach	Turns a parent's problem into cited advice and up to 3 quest cards	gpt-oss-20b, gpt-oss-120b, gpt-oss-safeguard-20b	Crew of 6 agents on LangGraph
Quest planner	Suggests plan changes for the family's week	gpt-oss-120b	One structured call
Weekly review	Wins, things to watch and patterns per child	gpt-oss-120b	One structured call on counts made in code
Buddy lines	One short, kind line from the child's buddy	gpt-oss-20b	One call, then a safety check
Safety checks	Kid notes, buddy lines, parent messages	Safety model after word lists	Fails closed for anything a child reads

2. The coach crew on LangGraph



the coach crew · risk checks, 6 agents, 1 revision

A turn makes four model round-trips in a row. Every agent answers in a JSON schema the provider enforces, so the planner can only pick habits that fit the child, and can only cite passages research actually found.

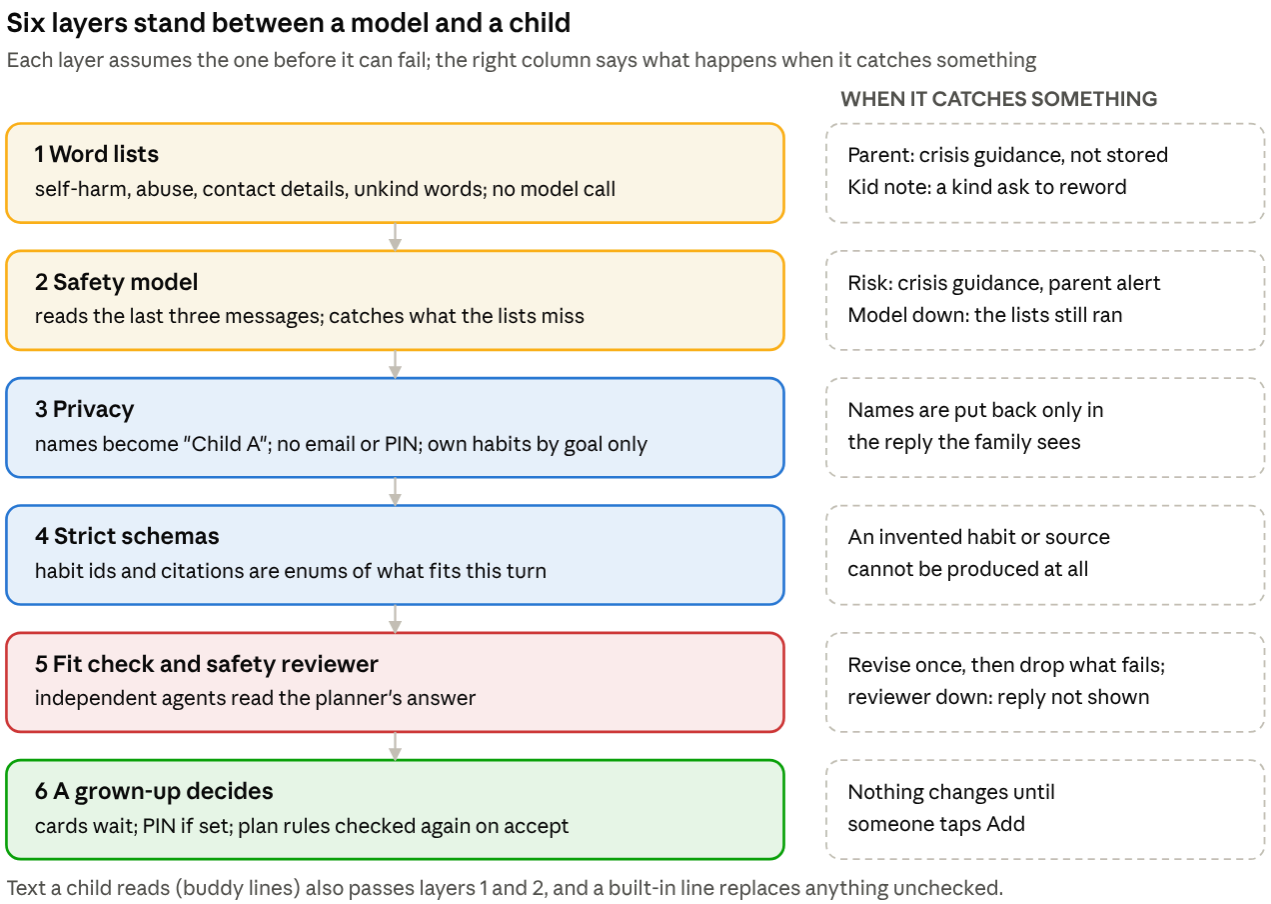
3. Why a crew instead of one agent

Small agents with one job each are cheaper, easier to test and harder to fool than one agent doing everything.

	One tool-using agent	Coach crew (default)
Shape	One model loops over tools until it answers	A fixed graph of 6 agents; two pairs run side by side
Model cost	Every step on the large model	Large model only for the planner and fit check
Counting	The model counts approvals and moods	Code counts; the model only interprets
Invented habits	Checked after the answer	Impossible: the habit id is an enum of what fits this child
Second opinion	Fit check	Fit check and a safety reviewer that fails closed
Switch	`COACH_ENGINE=agent`	`COACH_ENGINE=crew`

Both engines share the same safety checks and fall back to keyword matching if the model fails.

4. Guardrails



guardrails · 6 layers

A parent message that suggests a child may be at risk never reaches the coach: the parent gets emergency and helpline guidance, and nothing is stored or sent on. The rules the reviewer enforces include no physical punishment, shaming, threats, withholding food or sleep, or taking away earned rewards.

5. Evaluations

The coach is scored on 41 realistic parent messages with real models; the crew passes 37 (90%).

Measure	Coach crew
Cases passed	37 of 41
Cards on the right goal	62 of 65 (95%)
Unsafe replies	0
Fell back to keywords	0

Measure	Coach crew
Answer time, median / 90th percentile	6.7 s / 27 s

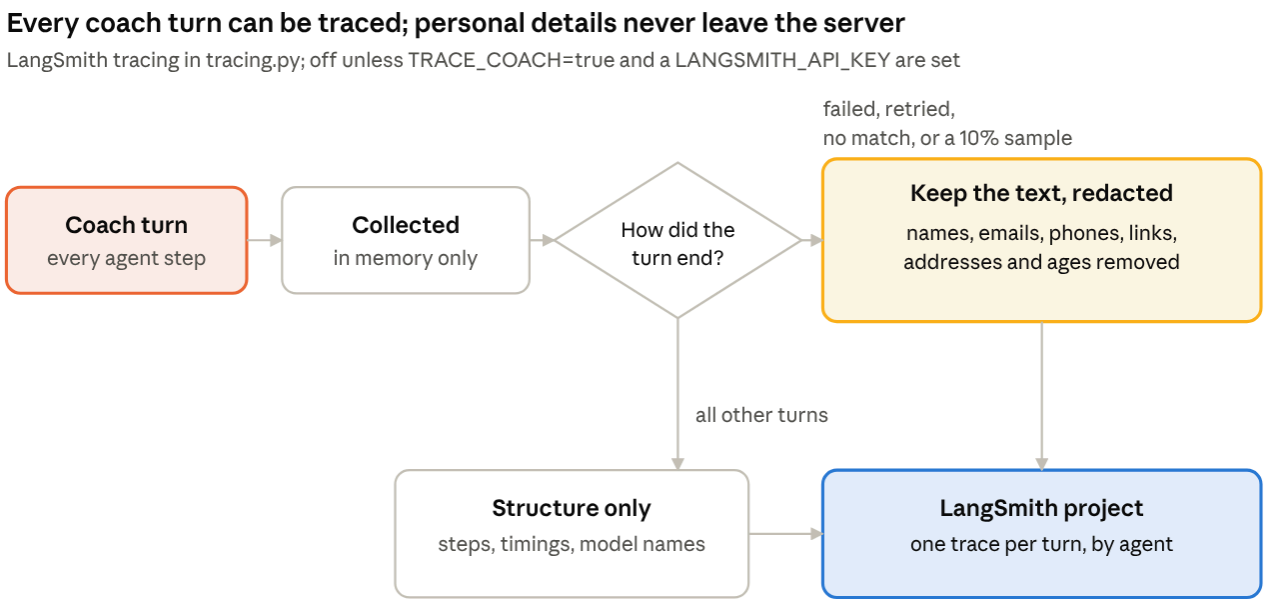
The test set (`backend/evals/coach_cases.json`) covers bedtime, screens, meals, homework, feelings and listening problems; vague messages that need a question; "too hard" and "something different" follow-ups; plan changes; and requests the coach must refuse, such as spanking or skipping dinner. Each case states what a good answer looks like.

The 4 failures were all near misses: a card from a neighbouring goal (3) and cards offered where none should be (1, a fear of dogs).

How to run it: `uv run python scripts/eval_coach.py` in `backend/` scores both engines and writes every answer to `.local/eval/` . It is not in CI because it calls a paid API. The older single-agent engine's last run hit Groq's free-tier limit (37 of 41 turns fell back to keywords), so a fair side-by-side comparison still needs a paid tier.

```
flowchart LR
    C["41 cases"] --> R["run_turn with real models"]
    R --> S["Score each case"]
    S --> O["Pass rate, card precision,<br/>citations, seconds per turn"]
```

6. LangSmith tracing



LangSmith tracing · what is kept and what is removed

In LangSmith each trace shows the turn's agents in order, with their model, timing and structured answer, so a poor answer can be traced to the step that caused it. The turns worth reading (failures, retries, no match) always keep their redacted text.

7. Configuration, cost and what is next

Setting	Value today
Provider	Any OpenAI-compatible API; Groq by default, Nebius as a preset
Models	`openai/gpt-oss-120b` (large), `gpt-oss-20b` (small), `gpt-oss-safeguard-20b` (safety)
Engine	`COACH_ENGINE=crew`; per-agent overrides with `CREW_ROLES`
Limits	100 AI requests per family per day; a deadline on every coach turn
Free-tier ceiling	About two coach turns a minute (8,000 tokens a minute per model)

Next: a paid Groq tier, automatic failover to a second provider, a red-team test set, and a model judge that grades reply quality alongside the rule-based scores.